Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model — qualifier level (Lista 7)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Notation (Benigno 2015). Lower-case letters are log deviations from the steady state; bars denote the long run. Households: $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$, $u = C^{1-\tilde{\sigma}^{-1}}/(1-\tilde{\sigma}^{-1})$, $v = L^{1+\eta}/(1+\eta)$, $\rho \equiv -\ln\beta$. Resources: $y = s_c c + g$, $\sigma \equiv \tilde{\sigma} s_c$. Firms: demand $Y(j) = (P(j)/P)^{-\theta}(C+G)$, technology $Y(j) = AL(j)$; a fraction α of prices is preset at P^e. Calibration unless stated: $\alpha = 0.75$, $\sigma = 0.5$, $\eta = 0.2$, $\theta = 8$, $\rho = 0.01$ (a quarter), $p^e = \bar{p} = 0$, $g = \bar{g} = 0$, no taxes.

Block 1 — Foundations: AD, AS, natural and efficient output, taxes

(Benigno §§1–5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) The slope κ of the AS curve $p - p^e = \kappa(y - y_n)$ under the calibration is:

A. 6.600

B. 0.733

C. 0.233

D. 2.200

b) (3 points) *AS–AD × taxation.* Productivity rises by $a = 0.02$ and a payroll tax raises the (log) markup by $\mu = 0.05$, both temporarily. The changes in $(y_n,\ y_e)$ are:

A. $(-0.0227,\ 0)$

B. $(0.0109,\ 0.0109)$

C. $(-0.0118,\ -0.0118)$

D. $(-0.0118,\ 0.0109)$

c) (2 points, T or F) Because θ sets the markup, a higher demand elasticity θ makes the AS curve steeper.

d) (2 points, T or F) *AS–AD × fiscal news.* Holding i, $\bar{p}$ and p^e fixed, news that long-run government spending will be higher ($\bar{g}\uparrow$, g unchanged) lowers current output.

Question 2. (15 points) Discursive. *AS–AD foundations × taxation × supply shocks.* Build the model from its primitives (no taxes unless stated).

a) (3 points) From the household problem with budget $PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$, derive the Euler equation and the labor-supply condition; log-linearize and combine with $y = s_c c + g$ to obtain the AD curve. Give its slope in the (y, p) plane, explain it, and list its shifters.

Solution:

b) (3 points) From the firm's problem derive the optimal flexible price, natural output and the AS curve with a fraction α of preset prices.

Solution:

c) (3 points) Derive efficient output from the planner's problem and compare it with natural output. Show how labor, payroll and consumption taxes affect each, and find the subsidy that makes the steady state efficient.

Solution:

d) (3 points) An oil shock raises the markup by $\mu = 0.05$ while productivity falls by $a = -0.02$, both for the short run only. Find the equilibrium output gap in terms of the natural real rate and the policy rate, compute the outcome if the bank keeps $i = \rho$ and the i that delivers $p = p^e$, and draw the AS–AD diagram (old and new AS, AD, both equilibria and y_e).

Solution:

e) (3 points) *Parameter analysis.* For the markup shock alone ($\mu = 0.05$, $a = 0$, $i = \rho$), how do the responses of output and prices depend on price rigidity? Find them in closed form as functions of κ, prove the signs, give the limits $\alpha \to 0$ and $\alpha \to 1$, fill the table for $\alpha \in \{0.5, 0.66, 0.75, 0.9\}$ and read it.

Solution:

Block 2 — Policy: optimal rules, the zero bound and household debt

(Benigno §§6–12 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Monetary × fiscal policy.* A temporary cut in consumption taxes of 2 points (like the 2009 IPI cut on cars, modeled as $\tau_c = -0.02$, entering the markup one for one) changes the natural real rate $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) + \sigma^{-1}(g - \bar{g}) + (\bar{\tau}_c - \tau_c)$ by:

A. $+2.00$ points

B. -1.82 points

C. $+0.18$ points

D. 0

b) (3 points) *Fiscal multipliers.* With i fixed, a temporary rise in g of 1% of GDP changes output and the gap $y - y_n$ by, respectively:

A. 1.000% and 0.000%

B. 0.909% and 0.000%

C. 0.732% and 0.067%

D. 0.976% and 0.067%

c) (2 points, T or F) At the zero lower bound with $r_n < 0$, a credible commitment to a long-run price level higher by $-r_n$ restores $y = y_n$ and $p = p^e$.

d) (2 points, T or F) *AS–AD × household debt.* In the borrower–saver economy with constant inflation between the short and the long run, more price flexibility reduces the output cost of a deleveraging shock.

Question 4. (15 points) Discursive. *Optimal policy × zero lower bound × household debt.*

a) (3 points) The bank minimizes Benigno's welfare loss $L = \frac{1}{2}(y - y_e)^2 + \frac{1}{2}\frac{\theta}{\kappa}(p - p^e)^2$ (eq. 33) subject to the AS. Find the optimal relation between the efficient gap and price surprises, describe it in the (y, p) plane and prove that shocks that leave μ unchanged create no trade-off.

Solution:

b) (3 points) For the shock of Question 2(d), compute the optimal y, $p - p^e$ and i and compare with strict price stability. Then show that no Taylor rule $i = \bar{\imath} + \psi_y(y - y_e) + \psi_\pi(p - p^e)$ with a constant intercept implements the optimum under productivity shocks ($a = \pm 0.004$).

Solution:

c) (3 points) Pessimism: long-run productivity is expected to fall, $\bar{a} = -0.05$, with $a = 0$. Show that the zero bound binds, compute output and prices at $i = 0$ with $\bar{p} = p^e$, find the $\bar{p}$ that restores the optimum, give the limits as $\kappa \to \infty$ and $\kappa \to 0$, and discuss credibility.

Solution:

d) (3 points) With borrowers and savers (share χ of savers, initial debt d_0), Benigno's AD is

$$y = \bar{y}_n - \varphi[i - (\bar{p}_n - p) - \rho] + \tfrac{1}{\chi}(g - \bar{g}) + \tfrac{1-\chi}{\chi}[\hat{d} + d_0(p - p^e)], \qquad \varphi \equiv \frac{\sigma\chi + (1-\chi)d_0\beta}{\chi},$$

where $\hat{d}$ is the change in the debt limit. Find dy/dp when $\bar{p}_n$ is anchored and when inflation $\bar{p}_n - p$ is constant, name the two channels, and derive the spending multipliers in both regimes; compute them with $\chi = 2/3$, $d_0 = 1.2$, $\beta = 0.98$.

Solution:

e) (3 points) *Parameter analysis.* How does the output response to a fall in the debt limit $\hat{d} = -0.1$ depend on price flexibility in each regime? Find it in closed form, prove the signs, give the stability bound, fill the table for $\alpha \in \{0.66, 0.75, 0.9\}$, and draw both responses against κ.

Solution: